

Least-squares

- ▶ least-squares (approximate) solution of overdetermined equations
- ▶ projection and orthogonality principle
- ▶ least-squares estimation
- ▶ BLUE property

Overdetermined linear equations

consider $y = Ax$ where $A \in \mathbb{R}^{m \times n}$ is (strictly) skinny, *i.e.*, $m > n$

- ▶ called *overdetermined* set of linear equations (more equations than unknowns)
- ▶ for most y , cannot solve for x

one approach to *approximately* solve $y = Ax$:

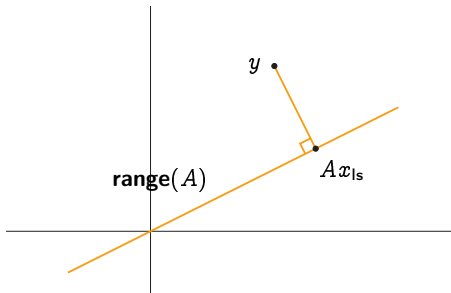
- ▶ define *residual* or error $r = Ax - y$
- ▶ find $x = x_{ls}$ that minimizes $\|r\|$

x_{ls} called *least-squares* (approximate) solution of $y = Ax$

Geometric interpretation

Given $y \in \mathbb{R}^m$, find $x \in \mathbb{R}^n$ to minimize $\|Ax - y\|$

Ax_{ls} is point in $\text{range}(A)$ closest to y (Ax_{ls} is *projection* of y onto $\text{range}(A)$)



Least-squares (approximate) solution

- ▶ assume A is full rank, skinny
- ▶ to find x_{ls} , we'll minimize norm of residual squared,

$$\|r\|^2 = x^T A^T A x - 2y^T A x + y^T y$$

- ▶ set gradient w.r.t. x to zero:

$$\nabla_x \|r\|^2 = 2A^T A x - 2A^T y = 0$$

- ▶ yields the *normal equations*: $A^T A x = A^T y$
- ▶ assumptions imply $A^T A$ invertible, so we have

$$x_{ls} = (A^T A)^{-1} A^T y$$

... a very famous formula

Least-squares (approximate) solution

- ▶ x_{ls} is linear function of y
- ▶ $x_{\text{ls}} = A^{-1}y$ if A is square
- ▶ x_{ls} solves $y = Ax_{\text{ls}}$ if $y \in \text{range}(A)$

Least-squares (approximate) solution

for A skinny and full rank, the *pseudo-inverse* of A is

$$A^\dagger = (A^\top A)^{-1} A^\top$$

- ▶ for A skinny and full rank, $A^\dagger$ is a *left inverse* of A

$$A^\dagger A = (A^\top A)^{-1} A^\top A = I$$

- ▶ if A is not skinny and full rank then $A^\dagger$ has a different definition

Projection on $\text{range}(A)$

Ax_{ls} is (by definition) the point in $\text{range}(A)$ that is closest to y , *i.e.*, it is the *projection* of y onto $\text{range}(A)$

$$Ax_{\text{ls}} = \mathcal{P}_{\text{range}(A)}(y)$$

- ▶ the projection function $\mathcal{P}_{\text{range}(A)}$ is linear, and given by

$$\mathcal{P}_{\text{range}(A)}(y) = Ax_{\text{ls}} = A(A^T A)^{-1} A^T y$$

- ▶ $A(A^T A)^{-1} A^T$ is called the *projection matrix* (associated with $\text{range}(A)$)

Orthogonality principle

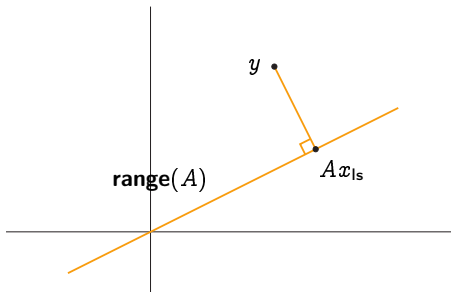
optimal residual

$$r = Ax_{ls} - y = (A(A^T A)^{-1} A^T - I)y$$

is orthogonal to **range**(A):

$$\langle r, Az \rangle = y^T (A(A^T A)^{-1} A^T - I)^T A z = 0$$

for all $z \in \mathbb{R}^n$



Completion of squares

since $r = Ax_{\text{ls}} - y \perp A(x - x_{\text{ls}})$ for any x , we have

$$\begin{aligned}\|Ax - y\|^2 &= \|(Ax_{\text{ls}} - y) + A(x - x_{\text{ls}})\|^2 \\ &= \|Ax_{\text{ls}} - y\|^2 + \|A(x - x_{\text{ls}})\|^2\end{aligned}$$

this shows that for $x \neq x_{\text{ls}}$, $\|Ax - y\| > \|Ax_{\text{ls}} - y\|$

Least-squares estimation

many applications in inversion, estimation, and reconstruction problems have form

$$y = Ax + v$$

- ▶ x is what we want to estimate or reconstruct
- ▶ y is our sensor measurement(s)
- ▶ v is an unknown *noise* or *measurement error* (assumed small)
- ▶ i th row of A characterizes i th sensor

Least-squares estimation

least-squares estimation: choose as estimate $\hat{x}$ that minimizes

$$\|A\hat{x} - y\|$$

i.e., deviation between

- ▶ what we actually observed (y), and
- ▶ what we would observe if $x = \hat{x}$, and there were no noise ($v = 0$)

least-squares estimate is just $\hat{x} = (A^T A)^{-1} A^T y$

BLUE property

suppose A full rank, skinny, and we have linear measurement with noise

$$y = Ax + v$$

consider a *linear estimator* of form $\hat{x} = By$

- ▶ B is called *unbiased* if $\hat{x} = x$ whenever $v = 0$
 - ▶ no estimation error when there is no noise
 - ▶ equivalent to left inverse property $BA = I$
- ▶ estimation error of unbiased linear estimator is

$$x - \hat{x} = x - B(Ax + v) = -Bv$$

- ▶ so we'd like B 'small' and $BA = I$

BLUE property

fact: $A^\dagger = (A^\top A)^{-1} A^\top$ is the *smallest* left inverse of A , in the following sense:

for any B with $BA = I$, we have

$$\sum_{i,j} B_{ij}^2 \geq \sum_{i,j} A_{ij}^{\dagger 2}$$

i.e., least-squares provides the *best linear unbiased estimator* (BLUE)